Bridging Expertise Gaps: A Comparative Study of AI Accessibility Across User Populations

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Abstract— **Conversational AI systems promise to democratize access to complex technical domains, yet designing for accessibility remains challenging when user populations span from complete novices to domain experts. This paper examines AI accessibility through two contrasting case studies: Skylar, an AI agent for cloud infrastructure operations targeting experienced engineers, and FinBot, a financial literacy chatbot designed for general audiences. Through technical system descriptions and an ongoing empirical study with middle school students (N=~25), I investigate how user expertise shapes interaction patterns, trust formation, and comprehension. Preliminary analysis suggests that accessibility requirements differ fundamentally across expertise levels, with implications for education, cloud operations, and other technical domains where AI serves as a knowledge bridge. This work contributes to human-AI interaction research by demonstrating that accessibility interventions must be calibrated to user background, with specific design implications for building AI systems that serve diverse populations.**

Keywords-component: Human-AI interaction, conversational AI, accessibility, expertise levels, trust formation, natural language interfaces

I. Introduction

When a cloud infrastructure monitoring system detects "high RDS latency due to suboptimal query execution patterns," an experienced engineer understands immediately: database queries are slow and need optimization. When a financial chatbot tells a teenager to "consider diversified portfolio allocation," the message is lost in translation. Both scenarios involve AI systems attempting to make technical information accessible, yet the accessibility challenges are fundamentally different. As AI systems proliferate across domains from healthcare to education, the question of accessibility becomes critical. But accessibility for whom? An expert may need transparency into AI decision-making logic, while a novice requires foundational context before any technical explanation makes sense. Designing AI that serves both populations and transitioning users along the expertise spectrum remains an open challenge in human-computer interaction research.

This paper addresses three interconnected questions:

RQ1: How do design requirements for conversational AI differ when serving expert versus novice users in technical domains?

RQ2: What patterns characterize novice users' questions, mental models, and misconceptions when interacting with domain-specific AI?

RQ3: What design principles enable AI systems to be accessible across the expertise spectrum?

A. Research Approach

I employ a comparative case study methodology, examining two AI systems I developed for contrasting user populations: Skylar (expert-focused cloud operations assistant) and FinBot (novice-focused financial literacy chatbot). This technical development is complemented by an ongoing empirical investigation with middle school students (ages 11-13) as representative novice users, using mixed-methods research to understand their interaction patterns, trust formation, and conceptual models of AI systems.

B. Contributions

This paper contributes:

- **Comparative system analysis:** Technical descriptions of two contrasting AI assistants with architectural rationale for divergent design choices
- **Empirical user study:** Investigation of middle school students' interaction patterns with domain-specific conversational AI (in progress, N=~25)
- **Question taxonomy:** Classification of user questions across expertise levels, revealing how information needs shift with domain knowledge
- **Design principles:** Evidence-based guidelines for building accessible AI systems serving diverse user populations

- **Interdisciplinary methodology:** Integration of technical system development with human subject's research and HCI analysis

II. RELATED WORK

A. Conversational AI and Natural Language Interfaces

Recent advances in large language models have enabled sophisticated conversational agents capable of handling complex domain-specific queries. However, most research focuses on task completion metrics rather than how diverse user populations understand and trust these systems. Existing work demonstrates that conversational interfaces can reduce barriers to technical information, but studies typically focus on homogeneous user populations rather than examining how expertise level moderates' interaction patterns.

B. AI Explainability and Trust

The explainable AI (XAI) literature emphasizes transparency in AI decision-making. However, existing work often assumes technical competence to interpret explanations. Studies on user trust in AI systems suggest trust formation depends on reliability, transparency, and predictability, but these studies rarely examine how expertise level moderates these factors. The gap between expert and novice trust requirements remains underexplored.

C. Domain-Specific AI Applications

AIOps research has explored AI for IT operations and cloud monitoring, while educational technology literature examines AI tutoring systems. Few studies bridge these domains to examine how the same AI principles apply across expertise levels or investigate comparative design requirements for serving diverse populations.

D. Human-AI Interaction Across User Populations

Research on expert-novice differences in technology interaction provides frameworks for understanding how domain knowledge shapes problem-solving approaches. Recent work on children's understanding of AI reveals developmental factors in AI conceptualization. This paper extends this literature by directly comparing systems designed for opposite ends of the expertise spectrum using mixed-methods empirical research.

E. Research Gap

While existing research has advanced conversational AI, explainability, and domain applications independently, few studies examine how to design for accessibility across the expertise spectrum using comparative empirical methods. This paper addresses that gap through parallel technical development and human subjects' research.

III. SYSTEM DESCRIPTIONS

A. Design Philosophy

Both Skylar and FinBot were designed with user-centered principles but optimized for different populations. The design philosophy prioritized contextual relevance over generic responses, domain accuracy over conversational breadth, and transparency appropriate to user expertise. However, implementation of these principles varied dramatically based on target user knowledge.

B. Skylar: AI Assistant for Cloud Operations

1) Motivation and Use Case

Cloud infrastructure complexity creates diagnostic challenges. When Amazon OpenSearch Service clusters display red or yellow health status, engineers must synthesize data from multiple sources (CloudWatch metrics, logs, configurations) to identify root causes. This manual process is time-consuming, prone to inconsistency, and creates dependency on specialized knowledge.

Target users: Cloud engineers, DevOps professionals, Site Reliability Engineers

Problem: Diagnosing infrastructure issues requires expertise and time

Solution: AI agent that automates L1 troubleshooting through conversational interface

2) Technical Architecture

Skylar's architecture comprises five integrated components:

Data Ingestion Layer: AWS Lambda functions collect real-time metrics from CloudWatch, configuration data from CloudFormation templates, and historical logs from S3, streaming to AWS Glue for ETL processing.

Analytics Layer: AWS Glue transforms raw data into structured formats and loads into Redshift. Custom SQL queries implement anomaly detection using statistical methods (z-score analysis, moving averages) comparing current metrics against historical baselines.

AI Reasoning Layer: When Skylar receives queries via Amazon Lex, it (a) parses user intent (e.g., "why is my app slow?" → investigate latency), (b) queries Redshift for relevant metrics and anomalies, (c) applies rule-based logic for common issues (CPU >80% + increased response time → resource constraint), and (d) generates hypothesis with supporting evidence.

Natural Language Generation: Responses use parameterized templates with specific data: "Your RDS instance is experiencing high CPU utilization (92%, up 40% from baseline). This correlates with increased query volume from [service-name]. Recommendation: Review slow query logs and consider read replica scaling."

Interface Layer: Web-based chat interface (Lex) and voice interface (Amazon Connect) with embedded visualizations (metric graphs, log snippets).

Backend Analytics Pipeline: All interactions log to CloudWatch, archive in S3, process through Glue crawlers, and visualize in QuickSight dashboards. This creates a data pipeline: interactions → insights → system improvements → faster resolution.

3) Design Rationale for Expert Users

Assumed knowledge: Uses technical terminology (RDS, latency, query execution) without explanation experts possess domain vocabulary.

Dense information: Provides multiple metrics, timestamps, correlations in single response experts can parse complex data efficiently.

Actionable specificity: Recommends exact remediation steps with technical parameters experts know how to execute.

Transparency over simplicity: Shows raw data (log lines, metric values) alongside interpretations experts want to verify AI reasoning.

Efficiency-focused: Minimizes conversational overhead, prioritizes diagnostic speed.

4) Performance Outcomes

Skylar reduced mean time to resolution by 40% for common cluster health issues. Analysis of interaction logs revealed 80% of yellow cluster issues stemmed from disk space shortages, enabling automatic prioritization of this diagnostic check.

C. FinBot: AI Assistant for Financial Literacy

1) Motivation and Use Case

Financial literacy gaps, particularly among youth and early-career professionals, stem partly from intimidating terminology and lack of accessible explanations.

Target users: Students, early-career professionals, anyone learning finance basics

Problem: Financial concepts use jargon that intimidates beginners making learning about these concepts complex

Solution: Conversational agent explaining financial concepts in accessible language

2) Technical Architecture

FinBot uses simpler architecture reflecting different requirements:

Knowledge Base: SQL database containing (a) 500+ curated financial Q&A pairs, (b) glossary with definitions at three reading levels (elementary, middle school, adult), (c) contextual examples and analogies, and (d) common misconceptions and clarifications.

NLP Processing: Python-based intent classifier using scikit-learn identifies query category (saving, investing, credit, budgeting) and extracts key entities requiring explanation.

Response Generation: Retrieval-based system that (a) matches user query to relevant knowledge base entry, (b) adjusts reading level (defaults to middle school), (c) includes analogy or example grounding abstract concepts, and (d) ends with follow-up question encouraging exploration.

Example interaction:

- Query: "What's compound interest?"
- Response: "Compound interest is when you earn interest on your interest like a snowball getting bigger as it rolls downhill, picking up more snow. If you save \$100 and earn \$5 interest, next time you'll earn interest on \$105, not just the original \$100. Want to see an example with real numbers? Or want to use my compound calculator"

3) Design Rationale for Novice Users

No assumed knowledge: Defines every term, even seemingly basic concepts.

Analogies and examples: Every abstract concept paired with concrete illustration. Has learning platform encouraging reading and watching financial videos.

Shorter responses: 2-3 sentences maximum before checking user understanding.

Encouraging tone: Phrases like "Great question!" reduce intimidation.

Progressive disclosure: Starts simple, offers detail only if user requests.

Conversational scaffolding: Suggests next questions to guide learning.

4) Technical Challenges

Key challenges include (a) maintaining simple language without condescension, (b) handling edge cases outside knowledge base, (c) determining when users need clarification versus more detail, and (d) avoiding information overwhelming.

D. Comparative Analysis

TABLE I shows design dimension comparisons between Skylar and FinBot:

TABLE I

DESIGN DIMENSION COMPARISON

	Design Dimension Comparison		
	Dimension	*Skylar(Expert)*	*FinBot(Novice)*
1.	**Target Users**	Cloud engineers	Students
2.	**Domain**	Cloud infrastructure	Personal finance
3.	**Assumed Knowledge**	High(AWS, Databases)	None
4.	**Vocabulary**	Technical jargon	Plain language
5.	**Response Length**	Dense	Brief
6.	**Interaction Style**	Efficiency-focused	Educational
7.	**Transparency**	Shows raw data	Abstracts detail
8.	**Primary Goal**	Diagnose quickly	Build understanding
9.	**Follow-up**	Assumes next steps	Suggests questions

Key Insight: "Accessibility" means different things for different users depending on expertise. For experts, accessibility means efficient access to detailed diagnostic data. For novices, accessibility means comprehensible explanations building foundational knowledge. A single system optimized for one population would fail the other.

IV. EMPIRICAL STUDY: MIDDLE SCHOOL USER INVESTIGATION

A. Motivation

While Skylar's design was informed by professional experience with cloud engineers, FinBot's novice-focused design required empirical validation. How do complete novices conceptualize AI? What questions do they ask? What makes AI responses helpful versus confusing? To investigate these questions, I am conducting a user study with middle school students.

B. Methodology

1) Participants

- **Sample:** N=~25 students from Dorothy Hamm Middle School computer science class
- **Demographics:** Ages 11-13
- **Prior AI experience:** [To be measured via pre-survey]
- **Recruitment:** Teacher invitation to existing class, voluntary participation

2) Procedure

The study consists of a 40-minute classroom session:

1. **Pre-survey** (3 min): Baseline AI familiarity and comfort
2. **Educational presentation** (20 min): Introduction to AI concepts, explanation of how AI systems work, live FinBot demonstration
3. **Interactive activity** (12 min): Small group discussions of AI interaction scenarios
4. **Post-survey** (3 min): Comfort levels, usability perceptions
5. **Optional drawing** (2 min): Visual representation of "what AI looks like"

Throughout, I collect observational field notes of student questions, reactions, and discussion content.

3) Data Collection Instruments

Pre-Survey Measures: (a) AI familiarity (multiple choice), (b) comfort asking AI questions (5-point Likert), (c) AI concerns/confusions (open-ended), and (d) desired AI use cases (open-ended).

Post-Survey Measures: (a) updated comfort level (5-point Likert), (b) FinBot usability perception (5-point scale), (c) key learning (open-ended), (d) suggested improvements (open-ended), and (e) emotional response (multiple choice).

Observational Data: Real-time questions during presentation, group discussion content, and reactions to FinBot demonstration.

Visual Data: Student drawings of AI (optional).

4) Analysis Approach

Quantitative: Paired t-tests comparing pre/post comfort scores; descriptive statistics for usability ratings and emotional responses.

Qualitative: Thematic coding of open-ended responses and observational notes using inductive approach. [Upon data collection, inter-rater reliability will be established through independent coding of 20% of data]

Visual: Content analysis of drawings, categorizing representation types (anthropomorphic, abstract, technological).

C. Preliminary Expectations

Based on pilot observations and related literature, I anticipate:

H1: Student comfort with AI will increase significantly from pre-test to post-test following educational intervention and hands-on interaction.

H2: Student questions will cluster into categories reflecting: (a) AI's cognitive nature ("Does AI think?"), (b) trust and reliability ("Can I trust AI?"), (c) practical capabilities ("What can AI do?"), and (d) ethical concerns ("Is AI dangerous?").

H3: Students will exhibit anthropomorphic mental models of AI, as evidenced by drawings depicting robot-like or human-like entities.

H4: Students will identify plain language, examples, and uncertainty acknowledgment as features making AI helpful; jargon and assumed knowledge as features making AI confusing.

D. Expected Findings

[This section will be completed after complete data collection and analysis. Expected content includes quantitative results (pre/post comfort level changes, FinBot usability ratings, emotional response distributions), qualitative findings (question taxonomy with categories and representative examples), misconceptions and mental models (common misconceptions, mental model patterns, implications for AI education), and usability insights (helpful vs. confusing features, suggested improvements).]

E. Comparison: Novice vs. Expert User Needs

TABLE II will show comparative findings upon completion:

TABLE II
USER NEED COMPARISON (TO BE COMPLETED)

	User Need Comparison		
	Dimension	***Middle School Students***	***Cloud Engineers***
1.	Primary Questions	TBD	“Why? How to fix?”
2.	Information Need	TBD	Diagnostic details
3.	Preferred Language	TBD	Technical precision
4.	Trust Basis	TBD	Accuracy, credibility

V. DISCUSSION

A. Design Principles for Accessible AI (Preliminary)

Based on system development and preliminary observations, I propose five design principles:

Principle 1: Know Your User and Commit. Attempting to serve both novices and experts with a single system often results in serving neither well. Skylar's technical language would intimidate students; FinBot's simplifications would frustrate engineers. Design choice: explicitly select target population and optimize for their needs OR implement adaptive interfaces adjusting based on detected user expertise.

Principle 2: Match Language to Mental Models. Novices lack domain-specific mental models they need plain language and analogies to build conceptual scaffolding. Experts have rich mental models they need precise terminology to activate existing knowledge.

Principle 3: Calibrate Information Density. [To be expanded upon completion of study with specific examples from student feedback.]

Principle 4: Make Uncertainty Explicit. Both populations need to know when AI is uncertain, but for different reasons. [To be expanded with study findings.]

Principle 5: Support Different Success Criteria. For students, success may mean "I learned something new." For engineers, success means "I solved the problem quickly." [To be expanded with empirical support.]

B. Implications for AI Design

For Educational AI: [To be completed based on middle school findings.] AI tutors should address student misconceptions about AI itself. Emotional engagement (curiosity, interest) may correlate with learning.

For Enterprise/Technical AI: Expert users need transparency more than simplicity. Efficiency matters respect expert time. Trust comes from accuracy and consistency.

For General-Purpose AI: Consider adaptive interfaces adjusting based on user behavior signals. Provide explicit "expert mode" and "beginner mode" options. Default to progressive disclosure.

C. Limitations

Current limitations: Study in progress, findings preliminary. Small sample size from single classroom. Single session (no longitudinal data). Lack of control group. Self-reported comfort may not reflect actual competence. Skylar user data based on observations rather than formal study.

Upon completion, additional limitations may include Generalizability across age groups, domain-specific findings, and cultural/linguistic factors not controlled.

D. Future Work

Immediate next steps: (a) complete data collection and analysis for middle school study, (b) conduct formal user study with cloud engineers using Skylar, and (c) expand participant pool across multiple schools and age groups.

Longer-term directions: (a) longitudinal study tracking how understanding and trust evolve over time, (b) adaptive system prototype adjusting language/density based on user signals, (c) cross-domain investigation (healthcare AI, legal AI), (d) intervention study testing whether addressing misconceptions improves interaction quality, and (e) investigation of individual differences within expertise levels.

VI. CONCLUSION

This paper investigates AI accessibility across user expertise levels through comparative case studies and empirical investigation. Through Skylar (expert-focused cloud operations assistant) and FinBot (novice-focused financial literacy chatbot), I demonstrate that "accessibility" is not monolithic design requirements differ fundamentally based on user expertise.

The ongoing empirical study with middle school students will provide data on how novices conceptualize AI, what questions they ask, and what features make AI interactions successful. Combined with observations of expert users, this research yields insights for building AI systems serving diverse populations.

As AI proliferates in education, healthcare, governance, and daily life, ensuring accessibility across expertise levels becomes critical. This work contributes methodological approaches (comparative system development + empirical user study) and preliminary design principles for creating AI that empowers rather than excludes users.

The question is not just "What can AI do?" but "How do humans with different backgrounds understand, trust, and work with AI?" Answering this requires bridging computer science, linguistics, cognitive science, and education the interdisciplinary approach central to this investigation.

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NOTE TO REVIEWERS:

This research paper is submitted as a draft, as permitted by the application guidelines. The empirical study described in Section IV is currently in the consent collection phase. While data collection has not yet been completed due to required parental consent procedures and school scheduling coordination, the draft demonstrates my research methodology, study design, and analytical approach. I have expanded the literature review (Section II) to include additional recent work in conversational AI, explainable AI, and human-AI interaction.

— Wiam Salih, January 2026

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